

Coupled Multiplicity Floer Complex (CMFC)

This section rewrites the proposal's equations **(1)–(5)** into a single, coordinate-free Floer-type definition that supports a well-posed chain complex.

0. Starting point: equations (1)–(5) as given

We begin from the five expressions in the note:

1. Extended operator

$$F(u) = \frac{\partial u}{\partial t} + J\nabla H(u) + \sum_{i,j} T_{ij} \cdot \nabla \Phi(u) + \xi(t). \quad (1)$$

2. Tensor/TQFT-style observable

$$\Psi(u) = \sum_{i,j} T_{ij} (u_i \otimes u_j) e^{i(\theta_i - \theta_j)}. \quad (2)$$

3. Euler characteristic refinement (as written)

$$\chi(M) = \int_M \left(R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 4R_{\mu\nu} R^{\mu\nu} + R^2 \right) d^4x. \quad (3)$$

4. Hybrid quantum–classical correction term

$$F(u) = \frac{\partial u}{\partial t} + J\nabla H(u) + \sum_{m,n} \gamma_{mn} u_m u_n \cos(\theta_m - \theta_n). \quad (4)$$

5. Adaptive/learning feedback functional

$$L(t) = \sum_{i=1}^N \nabla H(u_i) \cdot \cos(\omega_i t + \phi_i) \cdot F(t) \cdot (1 + \varepsilon_i(t)). \quad (5)$$

The goal is to reinterpret these in a standard Floer analytic framework (elliptic PDE on $\mathbb{R} \times S^1$) while preserving the intended ingredients: multiplicity, tensor-network couplings, stochastic feedback, and prime-based encoding.

1. Geometric setup and hypotheses

Let (M, ω) be a symplectic manifold. Fix one of the following standard settings (either is sufficient):

(A) Exact/convex-at-infinity setting

- $\omega = d\lambda$ is exact and M is convex at infinity (or geometrically bounded, as needed for Floer compactness).
- Hamiltonians are chosen admissibly at infinity (e.g. linear or compactly supported) so that Floer trajectories have compactness.

(B) Closed monotone setting

- M is closed and monotone: there exists $\kappa > 0$ such that $c_1(TM) = \kappa [\omega]$ on $\pi_2(M)$.
- Choose Novikov coefficients if needed to control bubbling.

Fix: - A smooth 1-periodic Hamiltonian $H_t: M \rightarrow \mathbb{R}$, $t \in S^1$. - A smooth 1-periodic family of ω -compatible almost complex structures J_t .

Multiplicity level

Fix an integer $N \geq 1$. We work on the product symplectic manifold

$$(M^N, \omega^{\oplus N}) \quad \text{with} \quad \omega^{\oplus N} = \omega \oplus \cdots \oplus \omega.$$

Write points as $\mathbf{x} = (x^1, \dots, x^N) \in M^N$, and maps as $\mathbf{u} = (u^1, \dots, u^N)$.

Coefficient ring and “prime encoding”

Let Λ be a Novikov-type coefficient ring appropriate to the setting (e.g. $\mathbb{Z}_2$ -Novikov in the monotone case; $\mathbb{Z}_2$ in simple exact cases). Define the **prime-extended ring**

$$\mathcal{R} := \Lambda[p_1^{\pm 1}, \dots, p_N^{\pm 1}],$$

where $p_1, \dots, p_N$ are formal prime generators. These variables record multiplicity/filtration data (they are *not* a literal identification of trajectories with primes).

Tensor coefficients T_{ij}

Let

$$T = (T_{ij})_{1 \leq i, j \leq N} \in \text{Mat}_N(\mathbb{R})$$

(or $\text{Mat}_N(\mathcal{R})$ if one wants algebraic weights as well). Analytically, taking $T_{ij} \in \mathbb{R}$ is the cleanest choice; algebraic prime weights can be handled separately via $\mathcal{R}$.

2. Interaction potential Φ and the mode-coupling term

To interpret $\sum_{i,j} T_{ij} \cdot \nabla \Phi(u)$ in (1) in a coordinate-free way, we fold it into a single smooth coupling Hamiltonian/potential on M^N .

Pairwise coupling potential

Choose smooth 1-periodic functions

$$\phi_{ij,t}: M \times M \rightarrow \mathbb{R}, \quad t \in S^1.$$

Define the **tensor-weighted coupling potential**

$$\Phi_t(\mathbf{x}) := \sum_{i,j=1}^N T_{ij} \phi_{ij,t}(x^i, x^j).$$

Assume Φ_t and its derivatives satisfy the standard growth/boundedness conditions required for Floer compactness in your chosen setting.

Self-interaction (replacement for “ ηu^2 ”)

The term “ ηu^2 ” is not coordinate-free for manifold-valued u . Replace it by a smooth self-potential: - choose $V: M \rightarrow \mathbb{R}$ smooth, - choose $\eta \in \mathbb{R}$, and add

$$\Phi_t^{\text{self}}(\mathbf{x}) = \eta \sum_{i=1}^N V(x^i).$$

Mode-coupling (incorporating equation (4))

To absorb the $\sum_{m,n} \gamma_{mn} u_m u_n \cos(\theta_m - \theta_n)$ term into the same framework, choose: - smooth “mode functions” $a_m: M \rightarrow \mathbb{R}$ for $m = 1, \dots, M_0$, - phases $\theta_m \in \mathbb{R}/2\pi\mathbb{Z}$, - coefficients $\gamma_{mn} \in \mathbb{R}$ (or $\mathcal{R}$).

Define

$$\Phi^{\text{mode}}(\mathbf{x}) := \sum_{m,n=1}^{M_0} \gamma_{mn} \left(\sum_{i=1}^N a_m(x^i) \right) \left(\sum_{j=1}^N a_n(x^j) \right) \cos(\theta_m - \theta_n).$$

This is a coordinate-free analogue of (4): it is a smooth scalar function on M^N built from “mode amplitudes” a_m .

Total coupling

Define the **total multiplicity coupling Hamiltonian**

$$K_t(\mathbf{x}) := \Phi_t(\mathbf{x}) + \Phi_t^{\text{self}}(\mathbf{x}) + \Phi^{\text{mode}}(\mathbf{x}).$$

3. The CMFC Floer operator (single consistent replacement for (1) and (4))

Let $H_t^{\oplus N}(\mathbf{x}) := \sum_{i=1}^N H_t(x^i)$. Let $X_{H_t^{\oplus N}}$ and X_{K_t} denote the Hamiltonian vector fields on $(M^N, \omega^{\oplus N})$.

Deterministic perturbation term $\xi(t)$

To keep the Floer equation as a (perturbed) gradient-flow equation of an action functional, assume the perturbation is Hamiltonian: - choose a smooth 1-periodic perturbation $G_t: M^N \rightarrow \mathbb{R}$, - set $\xi_t := X_{G_t}$.

(If you truly want stochastic forcing, treat G_t as a random process and interpret the resulting equation as an SPDE; that is a separate analytic package and is not required for the basic CMFC definition.)

CMFC operator on the Floer cylinder

Define the **Coupled Multiplicity Floer operator**

$$\mathcal{F}_{\text{CMFC}}(\mathbf{u}) := \partial_s \mathbf{u} + J_t(\mathbf{u}) \left(\partial_t \mathbf{u} - X_{H_t^{\oplus N}}(\mathbf{u}) - X_{K_t}(\mathbf{u}) - \xi_t(\mathbf{u}) \right)$$

for maps $\mathbf{u}: \mathbb{R} \times S^1 \rightarrow M^N$.

This is the Floer-elliptic replacement of (1): the term $J\nabla H$ is realized as $J(\cdot)(-X_H)$ via the standard identity between gradient and Hamiltonian vector fields under the metric $g(\cdot, \cdot) = \omega(\cdot, J\cdot)$, and the " $T_{ij} \cdot \nabla \Phi$ " contribution is encoded by the scalar coupling K_t built from T_{ij} and $\phi_{ij,t}$.

4. CMFC chain complex

Generators

Assume the 1-periodic orbits of the coupled Hamiltonian

$$\widetilde{H}_t := H_t^{\oplus N} + K_t + G_t$$

are nondegenerate. Let $\mathcal{P}(\widetilde{H})$ denote the set of its 1-periodic orbits in M^N .

Define the chain module

$$CF_*(\widetilde{H}, J; \mathcal{R}) := \bigoplus_{\mathbf{x} \in \mathcal{P}(\widetilde{H})} \mathcal{R} \cdot \mathbf{x}$$

graded by the usual Conley–Zehnder/Maslov index on M^N (with standard conventions depending on coefficients/orientations).

Prime weights

Fix homomorphisms

$$\lambda_i : \pi_2(M^N, \mathbf{x}^-, \mathbf{x}^+) \rightarrow \mathbb{Z}, \quad i = 1, \dots, N,$$

which encode the chosen “multiplicity data” (examples: component-wise symplectic area filtrations, intersection counts with chosen cycles, or interaction/event counts defined by a geometric rule). For a Floer trajectory class $[\mathbf{u}]$, define the prime monomial

$$\mathbf{w}([\mathbf{u}]) := \prod_{i=1}^N p_i^{\lambda_i([\mathbf{u}])} \in \mathcal{R}.$$

Differential

Let $\mathcal{M}_0(\mathbf{x}^-, \mathbf{x}^+)$ denote the moduli space of solutions $\mathbf{u}$ of $\mathcal{F}_{\text{CMFC}}(\mathbf{u}) = 0$ with Fredholm index 1, modulo $\mathbb{R}$ -translation in s , connecting $\mathbf{x}^-$ to $\mathbf{x}^+$.

Define

$$\partial \mathbf{x}^- := \sum_{\mathbf{x}^+ \in \mathcal{P}(\tilde{H})} \sum_{[\mathbf{u}] \in \mathcal{M}_0(\mathbf{x}^-, \mathbf{x}^+)} \# \mathcal{M}_0([\mathbf{u}]) \mathbf{w}([\mathbf{u}]) \mathbf{x}^+.$$

(Here $\# \mathcal{M}_0([\mathbf{u}])$ is the signed count when orientations are available; otherwise use $\mathbb{Z}_2$ -counts.)

Well-definedness assumption

Assume the usual Floer analytic package holds for $(M^N, \omega^{\oplus N})$ with $(\tilde{H}_t, J_t)$: - compactness up to breaking (and bubbling control in the monotone/Novikov setting), - transversality/regularity (e.g. by generic choices or abstract perturbations).

Then the standard boundary-of-1-manifold argument implies

$$\partial^2 = 0,$$

and we define the **CMFC homology**

$$HF_*^{\text{CMFC}}(M; N, T, \Phi, \gamma; \mathcal{R}) := H_*(CF_*(\tilde{H}, J; \mathcal{R}), \partial).$$

5. Tensor observable Ψ (equation (2) as a derived functional)

Equation (2) is interpreted as an **auxiliary observable** extracted from Floer data, not as part of the PDE definition.

Fix a finite-dimensional complex vector space V and a rule that associates to each loop $\mathbf{x}(t) \in M^N$ a vector $U(\mathbf{x}) \in V$, together with components u_i in a chosen basis and phases θ_i . (For numerical work, U can come from a feature map, Fourier expansion in a chart, or a spectral truncation.)

Define

$$\Psi(\mathbf{x}) := \sum_{i,j} T_{ij} (u_i(\mathbf{x}) \otimes u_j(\mathbf{x})) e^{i(\theta_i - \theta_j)} \in V \otimes V.$$

One may then define correlators/invariants by pushing Ψ through counts over moduli spaces (e.g. summing $\Psi(\mathbf{x}^+)$ weighted by Floer trajectory counts). This is compatible with CMFC provided Ψ depends only on the asymptotics (or is otherwise controlled).

6. Euler characteristic refinement (equation (3) made precise)

If M is a **closed oriented Riemannian 4-manifold**, the Chern–Gauss–Bonnet theorem gives

$$\chi(M) = \frac{1}{32\pi^2} \int_M (|\text{Riem}|^2 - 4|\text{Ric}|^2 + R^2) d\text{vol}.$$

In CMFC, $\chi(M)$ can be used as: - a scalar parameter weighting observables (e.g. rescaling correlators), or - an additional grading/filtration constant in the coefficient system.

It is **not** needed for $\partial^2 = 0$; it is an optional refinement layer.

7. Adaptive feedback / learning (equation (5) as parameter evolution)

Equation (5) is most naturally interpreted as defining a **control law** that evolves the Floer data $(T_{ij}, \gamma_{mn}, \phi_{ij,t}, G_t)$ over an external time τ (training time), while Floer theory runs on $(s, t) \in \mathbb{R} \times S^1$.

Feedback-driven family of CMFC data

Let

$$\mathcal{D}^\tau := (H_t, J_t, T^\tau, \gamma^\tau, \phi_{ij,t}^\tau, G_t^\tau)$$

be a piecewise-smooth family in τ , where the update rule is derived from the scalar functional

$$L^\tau(t) = \sum_{i=1}^N \nabla H(u_i^\tau) \cdot \cos(\omega_i t + \phi_i) \cdot F^\tau(t) \cdot (1 + \varepsilon_i^\tau(t)).$$

Assume $\tau \mapsto \mathcal{D}^\tau$ stays within the admissible class for Floer theory and avoids degeneracies except at isolated continuation events.

Continuation principle

Under standard hypotheses, a homotopy $\tau \in [0, 1]$ yields continuation maps

$$\Phi_{0 \rightarrow 1} : HF_*^{\text{CMFC}}(\mathcal{D}^0) \rightarrow HF_*^{\text{CMFC}}(\mathcal{D}^1),$$

which are isomorphisms when $\mathcal{D}^\tau$ is a regular admissible homotopy.

This is the mathematically clean way to incorporate “adaptive learning” into Floer-style invariants: learning updates produce a *path of CMFC complexes*, related by continuation.

8. Summary (single consistent CMFC definition)

CMFC is the Floer homology of the coupled Hamiltonian system on M^N defined by: - baseline Hamiltonian $H_t^{\oplus N}$, - tensor-weighted coupling potential Φ_t built from T_{ij} and kernels $\phi_{ij,t}$, - optional self and mode-coupling corrections (absorbing the intent of (4)), - Hamiltonian perturbation G_t representing deterministic noise $\xi(t)$, - prime-extended coefficients $\mathcal{R}$ encoding multiplicity filtrations.

The resulting operator is the elliptic Floer operator $\mathcal{F}_{\text{CMFC}}$ on $\mathbb{R} \times S^1$, and the differential counts its rigid solutions with prime weights.